

Week - 5

PART 1: ENVIRONMENTAL BIOLOGICAL ASPECTS

Topics Covered (VERY IMPORTANT for MCQ)

- Oxygen Demand → BOD, COD, TOC
- Oxygen in natural systems
- Associated SDGs
- Biogeochemical cycles → Carbon, Nitrogen, Phosphorus, Sulphur

 **MCQ Trap:** They may ask “which cycles are included?” → Answer: **C, N, P, S**

MAJOR ORGANISM GROUPS (IMPORTANT LIST)

- Viruses
- Bacteria
- Fungi
- Algae
- Protozoa
- Macrophytes (aquatic plants)
- Fish
- Rotifers
- Micro-crustaceans
- Macro-invertebrates

VERY IMPORTANT:

- Algae → used in biodiesel production
- Question can be: “Which organism is used for biofuel?” → Algae

OXYGEN DEMAND (CORE TOPIC)

Definition (MUST MEMORIZE WORDING)

BOD (Biochemical Oxygen Demand) =

 Amount of **dissolved oxygen** required by aerobic microorganisms

to **decompose organic matter** in water
at **specific temperature & time**

⚡ REACTIONS (VERY IMPORTANT)

1. Aerobic Decomposition

Organic matter + $O_2 \rightarrow CO_2 + H_2O + \text{New cells} + \text{Stable products}$

👉 Stable products: NO_3^- , PO_4^{3-} , SO_4^{2-}

2. Anaerobic Decomposition ⚠️ (VERY IMPORTANT DIFFERENCE)

Organic matter $\rightarrow CO_2 + H_2O + \text{New cells} + \text{Unstable products}$

👉 Unstable products:

- H_2S
- NH_3
- CH_4

⚠️ MCQ Trap:

- **Stable = aerobic**
- **Unstable = anaerobic**

🧠 SIMPLE BOD CONCEPT

- High BOD $\rightarrow$ **More oxygen consumed**
- Leads to $\rightarrow$ **Oxygen depletion**
- Result $\rightarrow$ **Aquatic life dies**

👉 MCQ: High BOD means?

✓ **Polluted water**

📊 DEFINITIONS & TYPES (VERY VERY IMPORTANT)

♦ BOD

- Oxygen used by microbes to oxidize **organic matter**

♦ CBOD (Carbonaceous BOD)

- BOD due to **carbon compounds only**
- ♦ **NBOD (Nitrogenous BOD)**
 - BOD due to **nitrogen compounds**
- ♦ **ThOD (Theoretical Oxygen Demand)**
 - Oxygen needed for **complete oxidation**
 - Assumes **all organic matter is biodegradable**
- ♦ **BOD₅ (MOST IMPORTANT)**
 - Measured over **5 days incubation**
 - Standard lab test
- ♦ **BOD_u (Ultimate BOD)**
 - Oxygen required for **complete oxidation**
- ♦ **COD (Chemical Oxygen Demand)**
 - Oxygen required using **chemical oxidant**
 - Faster than BOD

⚠ IMPORTANT RELATION:
 👉 **COD $\approx$ ThOD (should be near equal)**

BOD VALUES (VERY HIGH MCQ CHANCE)

Source	BOD ₅ (mg/L)
River	2
Domestic wastewater	200
Pulp & paper mill	400
Commercial laundry	2000

Sugar beet factory **10,000**

Tannery **15,000**

Brewery **25,000**

Cherry canning **55,000**

⚠ MCQ Tricks:

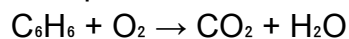
- **Clean water** → **low BOD (~2)**
- **Industrial waste** → **VERY HIGH BOD**

ThOD CALCULATION (IMPORTANT CONCEPT)

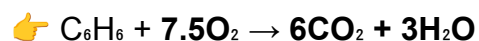
Step-wise:

1. Write oxidation reaction
2. Balance equation
3. Use stoichiometry

Example:



Balanced:



NUMERICAL (VERY IMPORTANT RESULT)

Given:

- 300 mg/L of C(H₂O)
- 50 mg/L of NH₃-N

Carbonaceous ThOD:

👉 **320 mg/L**

Nitrogenous ThOD:

👉 **229 mg/L**

✅ **Total ThOD:**

👉 **549 mg/L** 🔥 (VERY IMPORTANT FINAL VALUE)

⚠️ MCQ Trap:

- NH_3 given as **mg N/L** (not NH_3/L)

🧪 BOD EXPERIMENT (APPARATUS)

Important Items:

- BOD Incubator (20°C)
- Glass stoppered BOD bottles
- Pipette, burette, conical flask
- Measuring cylinder

🧴 CHEMICALS (REMEMBER KEY ONES)

- Potassium iodide (KI)
- Sodium thiosulphate
- Starch indicator
- Sulphuric acid
- Manganous sulphate
- Phosphate buffer
- Magnesium sulphate
- Calcium chloride
- Ferric chloride

⚠️ MCQ: These are used in **DO (Dissolved Oxygen) test**

🧪 PROCEDURE (STEP FLOW — IMPORTANT)

1. Add:
 - CaCl_2
 - MgSO_4
 - FeCl_3
 - Phosphate buffer
2. Prepare dilution:
 - **30 mL sample + 270 mL water**
3. Prepare:
 - **2 BOD bottles (sample)**

- **2 blank bottles**
- 4. Measure DO immediately
- 5. Incubate:
 - **5 days at 20°C**
- 6. Measure DO again

FORMULA (MOST IMPORTANT FOR NUMERICAL)

$$BOD_5 = DO_i - DO_f \quad P BOD_5 = \frac{D_{O_i} - D_{O_f}}{P} \quad BOD_5 = P(DO_i - DO_f)$$

Where:

- **DO_i** = Initial dissolved oxygen
- **DO_f** = Final dissolved oxygen
- **P** = Dilution factor

👉 Dilution factor:

= Volume of wastewater / (wastewater + dilution water)

ULTRA-IMPORTANT MCQ REVISION (LAST MINUTE)

- **BOD → Biological, COD → Chemical**
- **BOD₅ = 5 days, 20°C**
- **High BOD = polluted water**
- **Aerobic → stable products**
- **Anaerobic → CH₄, NH₃, H₂S**
- **COD ≈ ThOD**
- **River BOD ≈ 2 mg/L**
- **Max BOD → industrial waste**
- **Total ThOD example = 549 mg/L**

PART 2: BOD NUMERICALS + APPLICATIONS

EXAMPLE 1: RANGE OF BOD (VERY IMPORTANT)

Given:

- Sample = 10 mL
- Total volume = 300 mL
- Initial DO = 9 mg/L
- Required:
 - Minimum DO drop = 2 mg/L
 - Final DO $\geq$ 2 mg/L

Step 1: Dilution Factor

$$P = \frac{10}{300} \quad P = \frac{1}{30}$$

Step 2: Minimum BOD (for 2 mg/L drop)

$$BOD_5 = \frac{2}{10/300} = 60 \text{ mg/L} \quad BOD_5 = \frac{10}{300} \times 2 = 60 \text{ mg/L}$$

Step 3: Maximum BOD (so DO $\geq$ 2 mg/L)

$$DO_{\text{final}} = 2 \Rightarrow DO_{\text{drop}} = 9 - 2 = 7 \quad DO_{\text{final}} = 2 \Rightarrow DO_{\text{drop}} = 9 - 2 = 7$$

$$BOD_5 = \frac{7}{10/300} = 210 \text{ mg/L} \quad BOD_5 = \frac{10}{300} \times 7 = 210 \text{ mg/L}$$

FINAL ANSWER:

 BOD range = 60 to 210 mg/L

 MCQ Trap:

- Always check:
 - Minimum DO drop

- Minimum DO remaining

EXAMPLE 2: RAW SEWAGE BOD

Given:

- Sample = **2.5 mL**
- Total volume = **250 mL**
- Initial DO = **8 mg/L**
- Final DO = **5 mg/L**

Step 1: Dilution Factor

Dilution factor = $\frac{250}{2.5} = 100$
 $\text{Dilution factor} = \frac{250}{2.5} = 100$

Step 2: DO Depletion

$8 - 5 = 3 \text{ mg/L}$

Step 3: BOD

$\text{BOD} = 3 \times 100 = 300 \text{ mg/L}$

FINAL ANSWER:

 **BOD = 300 mg/L**

 Shortcut (VERY IMPORTANT):

 **BOD = DO depletion × Dilution factor**

EXAMPLE 3: 2% SOLUTION (VERY TRICKY)

Given:

- 2% solution → **Dilution factor = 100 / 2 = 50**
- DO depletion = **4 ppm = 4 mg/L**

BOD Calculation:

$\text{BOD} = 4 \times 50 = 200 \text{ mg/L}$

✓ **FINAL ANSWER:**

👉 **BOD = 200 mg/L**

⚠️ **MCQ Trap:**

- **% solution → convert to dilution factor**
- 1% → 100
- 2% → 50
- 10% → 10

🌍 **REAL-WORLD BOD INSIGHTS (VERY IMPORTANT THEORY)**

🌿 **GANGA RIVER DATA (CONCEPTUAL MCQ)**

- Some regions:
 - **< 3 mg/L → safe**
 - **3–10 mg/L → polluted**
 - **> 10 mg/L → highly polluted**

⚠️ **MCQ:**

- **Drinking water → very low BOD required**
- **High BOD → not suitable even for bathing**

🌍 **GLOBAL BOD PATTERNS**

- High pollution zones:
 - **India**
 - **China**
 - **Africa**
- Future (2050):
 - Pollution expected to **increase**

⚠️ **MCQ:**

- “Which regions have highest river pollution?” → **Developing regions**

👥 **POPULATION IMPACT (IMPORTANT)**

- Major polluted river basins:
 - **Ganga**
 - **Yangtze**
 - **Yellow River**
 - **Indus**

👉 Millions of people affected

⚖️ **DISCHARGE LIMITS (VERY IMPORTANT NUMBERS)**

👉 **BOD limit = 30 mg/L**

👉 **COD limit = 250 mg/L**

⚠️ MCQ Trap:

- They LOVE asking this directly

🧪 **DO TEST PROCEDURE (FLOW UNDERSTANDING)**

Key Steps:

1. Add reagents (KI, MnSO_4 etc.)
2. Add **sulfuric acid**
3. Formation of iodine
4. Titrate with **$\text{Na}_2\text{S}_2\text{O}_3$**
5. Use **starch indicator**
6. Endpoint → colorless

⚠️ MCQ:

- **Starch = indicator**
- **$\text{Na}_2\text{S}_2\text{O}_3$ = titrant**

💡 **ULTRA-IMPORTANT REVISION (MUST REMEMBER)**

- **$\text{BOD} = \text{DO depletion} \times \text{dilution factor}$**
- **BOD range question → min & max both**
- **2% solution → $\text{DF} = 50$**
- **BOD (sewage) ~ 300 mg/L**
- **Discharge limits → 30 (BOD), 250 (COD)**

- High BOD = polluted water
- DO must not fall below 2 mg/L

PART 3: ENVIRONMENTAL HEALTH + PATHOGENS

ENVIRONMENTAL HEALTH SCIENCE (DEFINITION)

Definition (VERY IMPORTANT)

 Study of **environmental factors affecting human health**

Includes:

- **Pollutants in air, water, soil**
- Transfer to humans via:
 - **Inhalation**
 - **Ingestion**
 - **Absorption**

 Leads to **adverse health effects**

 **MCQ Trap:**

- Mode of entry → all 3 options

BASIC REQUIREMENTS FOR HEALTHY ENVIRONMENT

- **Clean air, water, soil**
- **Safe & adequate food**
- **Stable global peaceful environment**

 **Direct MCQ expected**

ECOSYSTEM SERVICES (IMPORTANT CONCEPT)

Types:

1. **Provisioning**
 - Food, water, fuel
2. **Regulating**
 - Climate regulation
 - Water purification
 - Disease control

3. **Supporting**

- Nutrient cycling
- Soil formation

4. **Cultural**

- Recreational, aesthetic

Link to Human Well-being:

- **Security**
- **Health**
- **Basic material needs**
- **Good social relations**
- **Freedom of choice**

⚠ MCQ:

- “Which is NOT ecosystem service?” → tricky options

TARGET AREAS (VERY IMPORTANT LIST)

- **Water supply**
- **Wastewater treatment**
- **Waste management**
- **Land pollution control**
- **Vector control**
- **Food hygiene & safety**
- **Air quality management**
- **Radiation hazards**
- **Occupational health**
- **Noise management**

👉 Direct list-based MCQs expected

PATHOGENS IN WASTEWATER (VERY VERY IMPORTANT)

1. **BACTERIA**

- Present in faeces: $> 10^{10}$ / g
- Mostly **non-pathogenic**

When infection occurs:

- Pathogens increase
- Spread via faeces

👉 Common disease:

- **Diarrhea (most common)**
- **Cholera (most severe)**

2. 🦠 VIRUSES

- Present in faeces: $> 10^9$ / g

Important groups:

- **Adenoviruses**
- **Enteroviruses (Polio)**
- **Hepatitis A**
- **Reoviruses**
- **Rotavirus (VERY IMPORTANT)**

⚠️ MCQ:

- “Most common diarrhoea virus?” → **Rotavirus**

3. 🦠 PROTOZOA

- Cause:
 - **Diarrhea**
 - **Dysentery**
- Transmission:
 - Via **cysts**

Important pathogens:

- **Giardia lamblia**
- **Balantidium coli**
- **Entamoeba histolytica**

⚠️ MCQ:

- Only **3** protozoa considered major pathogens

4. HELMINTHS (WORMS)

- Parasitic worms
- Spread via:
 - **Eggs / larvae in excreta**

 Cause serious illness



WATER BORNE DISEASES (VERY IMPORTANT TABLE)



Viral:

- Hepatitis A
- Hepatitis E
- Rotavirus diarrhea



Bacterial:

- Typhoid
- Paratyphoid
- Cholera
- Bacillary dysentery
- E. coli diarrhea



Protozoal:

- Amoebiasis
- Giardiasis



Helminthic:

- Roundworm
- Threadworm
- Hydatid disease



MCQ:

- Match disease with organism type



MICROORGANISMS IN WASTEWATER (VISUAL CONCEPT)

Common:

- **Salmonella**
- **Shigella**
- **Vibrio cholerae**
- **Rotavirus**
- **Giardia**
- **Ascaris (worm)**

👉 High concentration in wastewater

SDG GOAL (VERY IMPORTANT)

 **Goal 6:**

👉 **Clean Water and Sanitation**

- Ensure **availability**
- Ensure **sustainable management**

⚠️ MCQ:

- SDG 6 = Water & sanitation

GLOBAL HEALTH FACTS (VERY HIGH MCQ CHANCE)

- **1.5 billion people lack sanitation**
- **1 in 4 health facilities lack basic water**
- **Millions of women give birth without clean water**
- **1 million+ deaths/year due to unclean births**

CHOLERA (IMPORTANT)

- Severe **diarrheal disease**
- Can **kill within hours**
- **Easily treatable**

👉 MCQ favorite

SANITATION STATUS (GLOBAL)

Types:

- **Improved sanitation**
- **Shared sanitation**
- **Unimproved sanitation**
- **Open defecation**

 **MCQ:**

- “Which is worst?” → **Open defecation**

ULTRA-IMPORTANT REVISION

- **Pollutants enter via inhalation, ingestion, absorption**
- **Rotavirus → diarrhea**
- **Cholera → bacterial**
- **Giardia → protozoa**
- **Helminths → worms**
- **SDG 6 → water & sanitation**
- **$>10^{10}$ bacteria in faeces**
- **Target areas → water, waste, air, noise etc.**

PART 4: WATER STRESS + TREATMENT METHODS

WATER STRESS (VERY IMPORTANT)

 Concept:

👉 Population living where **freshwater withdrawal > 40% of renewable resources**

Key Data (MEMORIZE RELATIVE VALUES)

- Asia & Pacific → Highest (~2964 million projected)
- Africa → Moderate (~552 million)
- Europe → ~239 million
- West Asia → ~119 million
- Latin America → ~137 million
- North America → ~133 million

⚠ MCQ:

- Highest water stress → **Asia**

SEWAGE TREATMENT (CASE STUDY)

Jakarta vs Sydney

Jakarta:

- Sewage generated: **1.3 million m³**
- Treated: **ONLY ~3%**

Sydney:

- Sewage generated: **1.2 million m³**
- Treated: **~100%**

⚠ MCQ:

- Developing cities → **low treatment**
- Developed cities → **high treatment**

GLOBAL WASTEWATER TREATMENT

- Developed regions:
 - **High treatment**
- Developing regions:
 - **High untreated wastewater**

 MCQ:

- “Which region has more untreated wastewater?” → **Developing countries**

BIOLOGICAL METHODS OF WASTE TREATMENT

 Important Methods:

- **Composting**
- **Anaerobic digestion**
- **Microbial fuel cell (MFC)**
- **Bioremediation**

COMPOSTING (VERY IMPORTANT)

 Definition:

 Conversion of **organic waste** → **soil-like material**

 Agents:

- **Microorganisms (bacteria, fungi)**
- **Invertebrates (earthworms, insects)**

 Process:

- Needs:
 - **Oxygen**
- Produces:
 - **Heat**
 - **CO₂**
 - **Water**

✓ Benefits:

- Reduces waste
- Produces **soil fertilizer**
- Improves **soil fertility**
- Reduces need for **chemical fertilizers**
- Improves **drought resistance**

⚠ MCQ:

- Composting → **aerobic process**

♻ **ANAEROBIC DIGESTION (VERY IMPORTANT)**

✓ Definition:

👉 Breakdown of organic matter **without oxygen**

💡 Steps (VERY IMPORTANT ORDER)

1. **Hydrolysis**
2. **Acidogenesis**
3. **Acetogenesis**
4. **Methanogenesis**

⚠ MCQ:

- Last step → **Methanogenesis**

🔥 Products:

- **Biogas (CH_4 , CO_2 , H_2S)**
- **Digestate (fertilizer)**

⚡ Applications:

- **Wastewater treatment**
- **Power generation**
- **Cooking gas**
- **Vehicle fuel**
- **Fertilizer production**

MICROBIAL FUEL CELL (MFC)

Definition:

 Bio-electrochemical system using **bacteria to generate electricity**

Working:

- Bacteria break organic matter → release:
 - **Electrons**
 - **Protons**
- Produces:
 - **Electric current**
 - **Hydrogen gas**

Benefits (VERY IMPORTANT)

Energy:

- Direct electricity
- No aeration needed

Environmental:

- **Low carbon footprint**
- **Less sludge**

Economic:

- **Low operating cost**
- **Energy recovery**

Operational:

- **Self-regeneration**
- **Stress resistant**

MCQ:

- MFC produces → **electricity**
- Anaerobic digestion → **biogas**

BIOREMEDIATION

👉 Use of microorganisms to clean pollutants

⚠️ MCQ:

- Used for **pollution cleanup**

ULTRA-IMPORTANT FINAL REVISION

🔥 **Must Remember:**

- **Water stress = >40% withdrawal**
- **Asia → highest stress**
- **Jakarta → 3% treated**
- **Sydney → ~100% treated**
- **Composting → aerobic**
- **Anaerobic digestion → no oxygen**
- **Methanogenesis → final step**
- **Biogas = CH₄**
- **MFC → electricity**
- **Bioremediation → pollution cleanup**

FINAL EXAM STRATEGY

If MCQ comes:

✓ **Match type questions**

- Composting vs anaerobic
- Bacteria vs virus vs protozoa

✓ **Data-based questions**

- 3% vs 100%
- 40% threshold

✓ **Process sequence**

- Hydrolysis → Methanogenesis

✓ **Application-based** Biogas vs electricity

📖 PART 5: NUTRIENT CYCLES (VERY IMPORTANT)

🔄 ECOSYSTEM FUNCTION

✅ Key Idea:

👉 Nutrients are recycled within ecosystem

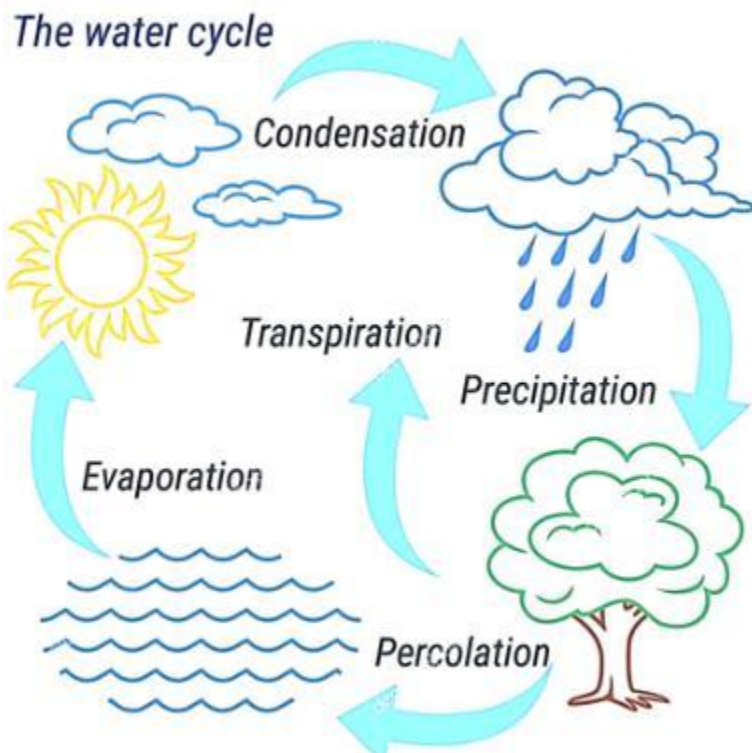
Cycles included:

- Carbon cycle
- Nitrogen cycle
- Phosphorus cycle
- Water cycle
- Oxygen cycle

⚠️ MCQ:

- “Which is NOT a nutrient cycle?” → tricky options

💧 WATER CYCLE (VERY DIRECT MCQ)



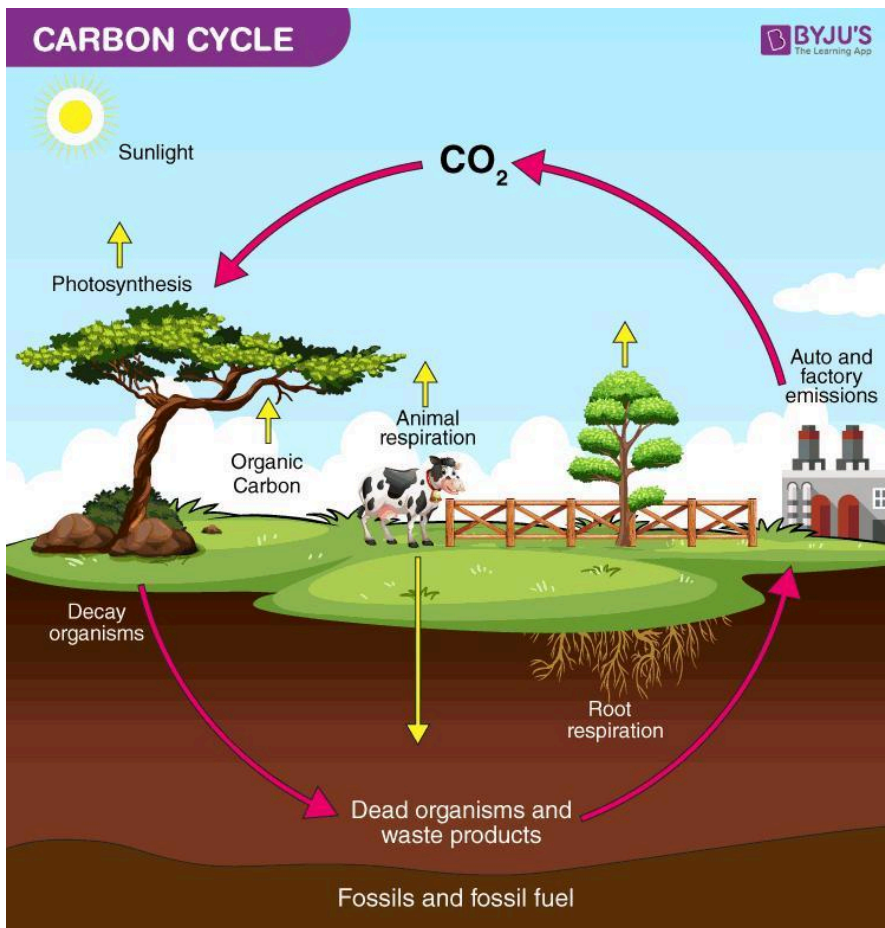
🔥 Main Processes:

- Evaporation
- Condensation
- Infiltration
- Surface runoff

⚠️ MCQ:

- Infiltration = water enters **soil**
 - Runoff = water flows on **surface**
-

🌿 CARBON CYCLE (VERY IMPORTANT)



✅ Definition:

👉 Movement of carbon among:

- **Biosphere**
- **Atmosphere**
- **Hydrosphere**
- **Geosphere**

Key Processes:

- **Photosynthesis** → **CO₂ absorbed**
- **Respiration** → **CO₂ released**
- **Decomposition**
- **Fossil fuel burning**

Important Concept:

- Carbon moves in **billions of tons/year**
- Human contribution → **red values (in diagram)**

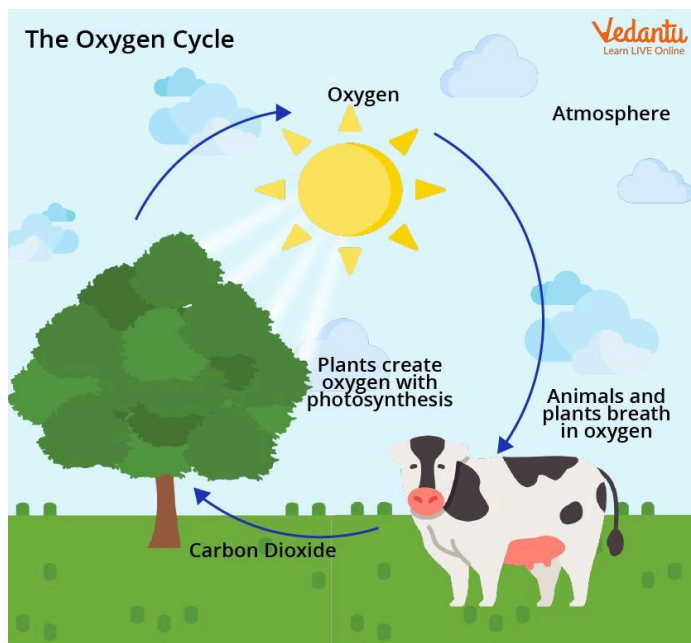
CARBON SINKS (VERY IMPORTANT)

- **Lithosphere (largest)** → **limestone**
- **Hydrosphere (2nd largest)** → **ocean**
- **Atmosphere** → **CO₂**
- **Biosphere** → **plants, animals**

MCQ:

- Largest carbon reservoir → **Lithosphere**

OXYGEN CYCLE



✓ Definition:

👉 Movement of oxygen between:

- Different chemical forms (ions, oxides, molecules)
- Across Earth systems

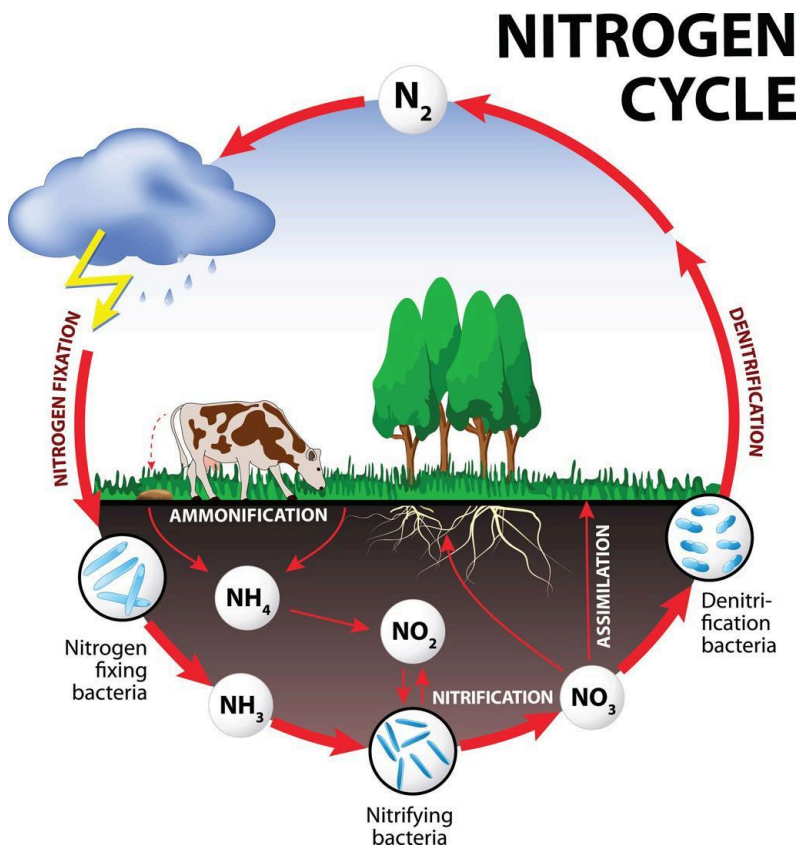
🔥 Key Points:

- Plants release O_2 (photosynthesis)
- Animals use O_2 (respiration)

⚠ Threats:

- Deforestation
- Ocean pollution

🌱 NITROGEN CYCLE (VERY VERY IMPORTANT)



6

✓ Definition:

👉 Nitrogen converts into multiple forms across ecosystems

🔥 Forms of Nitrogen:

- Urea $\rightarrow CO(NH_2)_2$
- Ammonia $\rightarrow NH_3$
- Ammonium $\rightarrow NH_4^+$
- Nitrite $\rightarrow NO_2^-$
- Nitrate $\rightarrow NO_3^-$
- Atmospheric N_2

⚠️ VERY IMPORTANT FACTS:

- 78% of atmosphere = N_2
- Cannot be used directly $\rightarrow$ must be **fixed**

KEY PROCESSES (MUST MEMORIZE)

1. Nitrogen Fixation

- $\text{N}_2 \rightarrow \text{NH}_3$
- Done by **bacteria (legumes)**

2. Ammonification

- Organic waste $\rightarrow \text{NH}_3$

3. Nitrification

- $\text{NH}_3 \rightarrow \text{NO}_2^- \rightarrow \text{NO}_3^-$

4. Assimilation

- Plants absorb nitrogen

5. Denitrification

- $\text{NO}_3^- \rightarrow \text{N}_2$ (back to atmosphere)



MCQ Trap:

- Order may NOT be fixed (complex cycle)



SPECIAL CONCEPT: TRIPLE BOND



N_2 has **strong triple bond**



Makes it **inert (like noble gas)**



That's why fixation is needed



ULTRA-IMPORTANT FINAL REVISION

- **Water cycle** $\rightarrow$ evap, cond, infiltration, runoff
- **Carbon sinks** $\rightarrow$ lithosphere largest
- **Photosynthesis** $\downarrow \text{CO}_2$
- **Respiration** $\uparrow \text{CO}_2$
- **Oxygen from plants**
- **Nitrogen = 78% atmosphere**
- **Fixation** $\rightarrow \text{N}_2 \rightarrow \text{NH}_3$

- **Nitrification** $\rightarrow \text{NH}_3 \rightarrow \text{NO}_3^-$
- **Denitrification** $\rightarrow \text{NO}_3^- \rightarrow \text{N}_2$
- **Triple bond** $\rightarrow$ inert nitrogen

PART 6: NITROGEN FIXATION + HUMAN IMPACT + PHOSPHORUS

HOW CAN WE USE N_2 ? (VERY IMPORTANT)

 N_2 gas cannot be used directly by plants/animals

 Must be converted into:

- Ammonium (NH_4^+)
- Nitrate (NO_3^-)

 MCQ:

- “Usable nitrogen forms?” $\rightarrow NH_4^+, NO_3^-$

NITROGEN FIXATION (CORE CONCEPT)

 Definition:

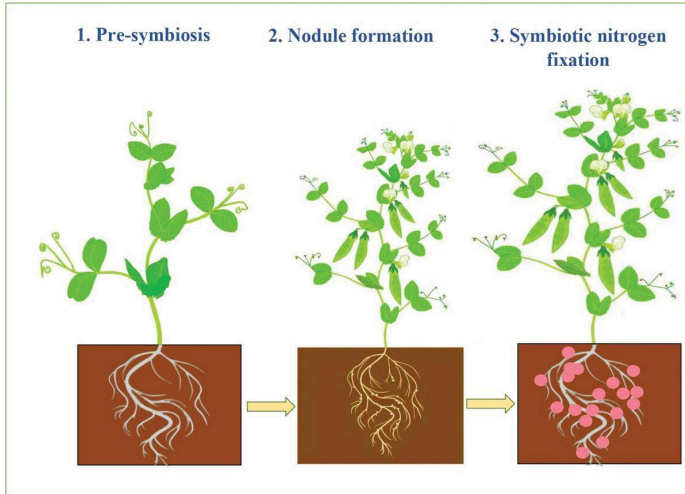
 Conversion of $N_2 \rightarrow NH_3 / NH_4^+$

 Key Reaction:

 $N_2 \rightarrow NH_3$ or NH_4^+

TYPES OF NITROGEN FIXATION

1. BIOLOGICAL FIXATION (VERY IMPORTANT)



Done by:

- Nitrogen-fixing bacteria (Rhizobium)

Key Points:

- Live in root nodules
- Found in legumes (beans, peas, clover)
- Symbiotic relationship

⚠ MCQ:

- “Where are nitrogen-fixing bacteria found?” → Root nodules

2. ⚡ ENVIRONMENTAL FIXATION

Caused by:

- Lightning
- Forest fires
- Volcanic/lava activity

👉 Breaks strong N_2 bond

3. 🏭 ANTHROPOGENIC FIXATION

Human activities:

- Fertilizers (synthetic nitrogen)

- Fossil fuel burning
- Legume cultivation

⚠ MCQ:

- Anthropogenic = human-induced

↺ NITROGEN MINERALIZATION (AMMONIFICATION)

✓ Definition:

👉 Organic nitrogen $\rightarrow$ NH_4^+

🔥 Done by:

- Decomposers (bacteria, fungi)

Key Sources:

- Dead organisms
- Manure

⚠ MCQ:

- Opposite = Immobilization

↺ NITRIFICATION & DENITRIFICATION

🔥 NITRIFICATION:

👉 $\text{NH}_3 \rightarrow \text{NO}_2^- \rightarrow \text{NO}_3^-$

🔥 DENITRIFICATION:

👉 $\text{NO}_3^- \rightarrow \text{N}_2$ (returns to atmosphere)

⚠ MCQ:

- Nitrification = oxidation
- Denitrification = reduction

HUMAN IMPACT (VERY IMPORTANT)

Fertilizers Effect:

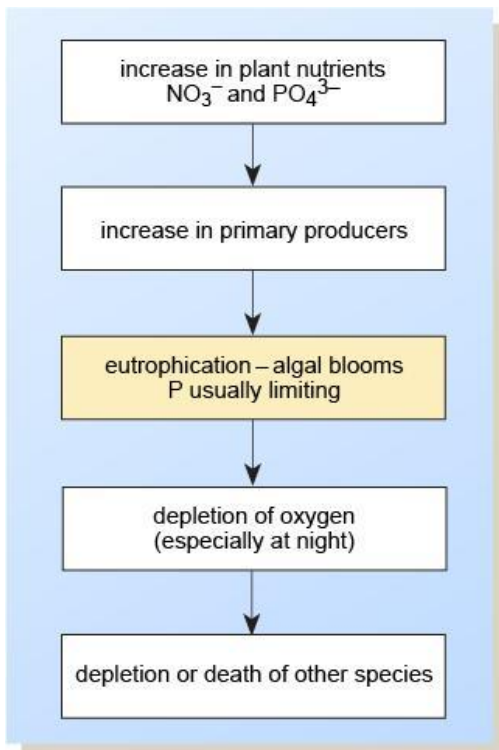
Runoff causes:

- Water contamination
- Groundwater pollution

Health Effects:

- Cancer (long-term)
- Respiratory distress (infants)

EUTROPHICATION (VERY IMPORTANT)



✓ Definition:

👉 Excess nutrients → over-enrichment of water

🔥 Effects:

- Algal blooms
- Fish kills
- Ecosystem imbalance

⚠ MCQ:

- Cause → Excess nitrogen/phosphorus

NITROGEN IN ATMOSPHERE (VERY TRICKY)

Forms:

- NO (smog)
- N₂O (greenhouse gas)
- NO_x (acid rain)

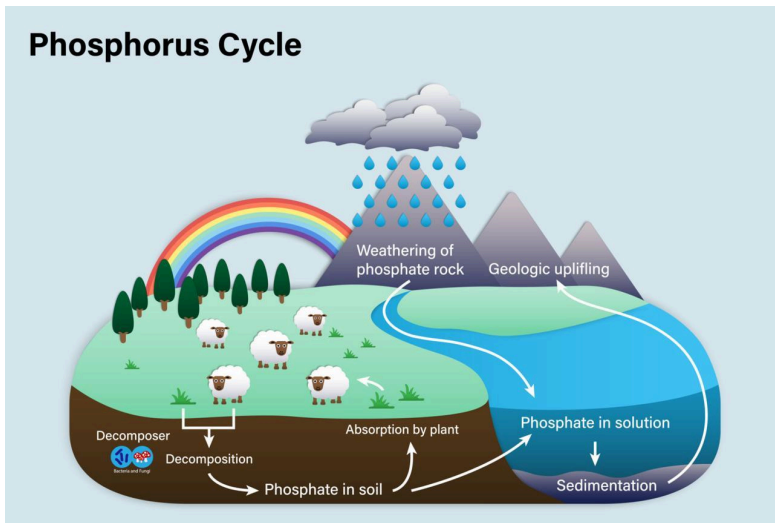
🔥 Nitrous Oxide (N₂O)

- Lifetime: ~110 years
- Depletes ozone
- Strong greenhouse gas

⚠ VERY IMPORTANT:

👉 GWP = 265–298 times CO₂

🪨 PHOSPHORUS CYCLE (VERY IMPORTANT)



✅ Definition:

👉 Movement of phosphorus in:

- Lithosphere
- Hydrosphere
- Biosphere

⚠️ VERY IMPORTANT DIFFERENCE:

👉 NO atmospheric phase

🔥 Key Processes:

- Weathering of rocks
- Uptake by plants
- Movement in food chain
- Sedimentation

⚠️ MCQ:

- “Which cycle has no gas phase?” → Phosphorus cycle

FERTILIZER DATA (CONCEPTUAL MCQ)

Trends:

- Fertilizer use is increasing globally
- Highest usage:
 - Asia

MCQ:

- “Region with highest fertilizer consumption?” → Asia

ULTRA-IMPORTANT FINAL REVISION

- $\text{N}_2 \rightarrow \text{NH}_3$ (fixation)
- Biological → root nodules
- Lightning → environmental fixation
- Fertilizers → anthropogenic
- Ammonification → NH_4^+
- Nitrification → NO_3^-
- Denitrification → N_2
- Eutrophication → algal bloom
- $\text{N}_2\text{O} \rightarrow 265\text{--}298\times \text{CO}_2$
- Phosphorus → NO atmosphere phase

PART 7: FINAL TOPICS (EUTROPHICATION + SULFUR + REMEDIATION)

FERTILIZER CONSUMPTION (IMPORTANT TREND)

Key Observations:

- Fertilizer use is continuously increasing
- Asia = highest consumption
- Africa = very low usage

MCQ:

- “Region with highest fertilizer use?” → Asia

FEAST OR FAMINE (CONCEPT)

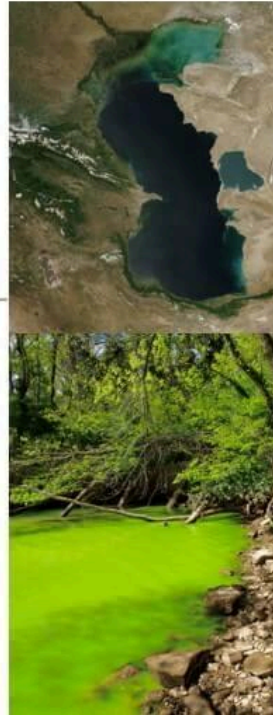
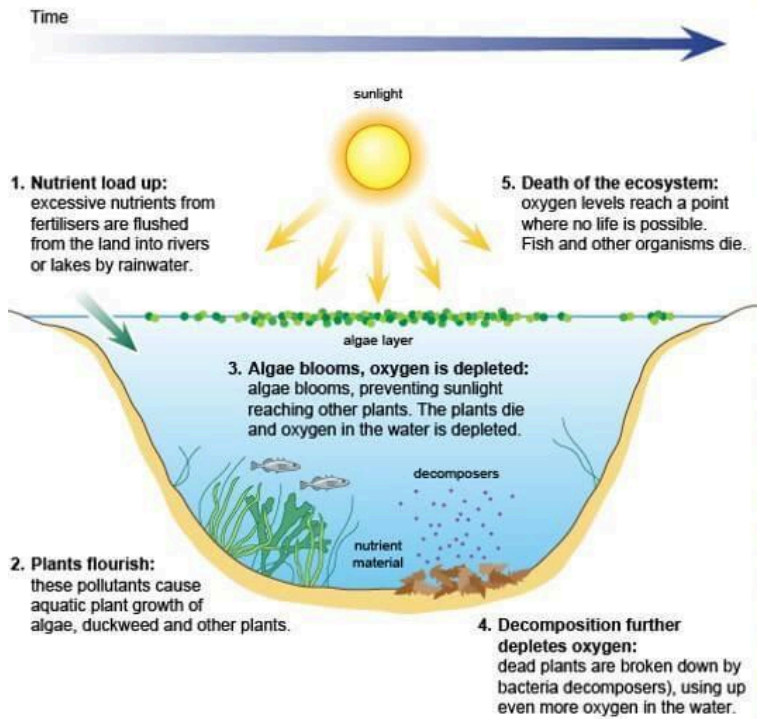
Soil fertility depends on nitrogen availability

- Too little nitrogen → low crop yield
- Too much nitrogen → environmental damage

MCQ:

- Balance of nutrients is important

EUTROPHICATION (REVISION + ADVANCED)



🔥 Process Flow:

1. Nutrients (N, P) increase
2. Phytoplankton growth ↑
3. Algal bloom
4. Algae die
5. Decomposition consumes oxygen
6. Oxygen depletion
7. Fish die

! Final Effects:

- Loss of:
 - Food
 - Habitat
 - Oxygen production

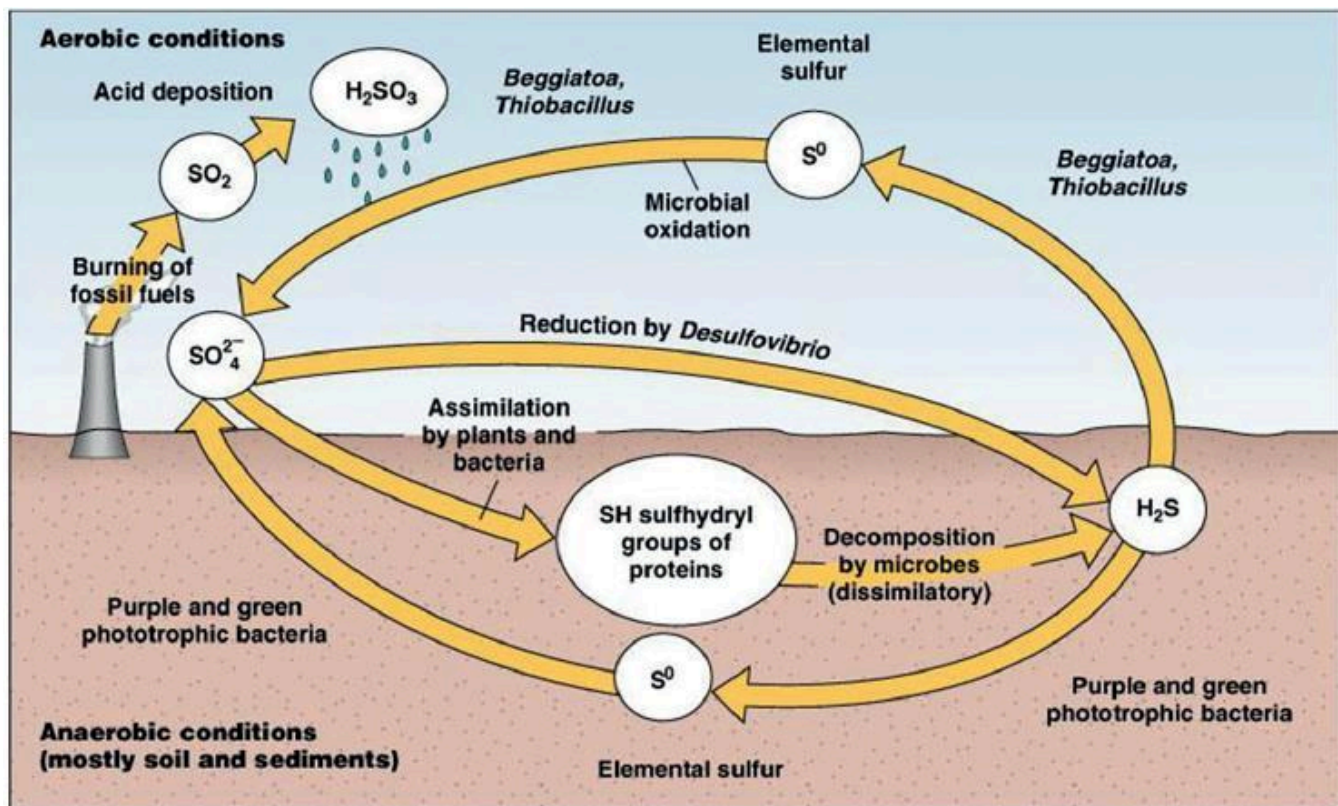
MEASURES TO REDUCE EUTROPHICATION (VERY IMPORTANT LIST)

1. Reduce phosphates in detergents
2. Reduce nitrate fertilizers
3. Tertiary sewage treatment
4. Divert waste away from lakes
5. Aeration of water bodies
6. Remove phosphate-rich plants
7. Remove sediments (dredging)

! MCQ:

- “Best method?” → Tertiary treatment

🌋 SULFUR CYCLE (IMPORTANT)



✓ **Definition:**

👉 **Movement of sulfur between:**

- **Rocks**
- **Water**
- **Living systems**

🔥 **Key Points:**

- **Sulfur released by:**
 - **Rock weathering**
 - **Volcanic eruptions**
 - **Decomposers (H_2S)**

⚠️ **VERY IMPORTANT:**

- **H_2S → converted to sulfur by bacteria**
- **SO_2 (industry) → H_2SO_4 (acid rain)**

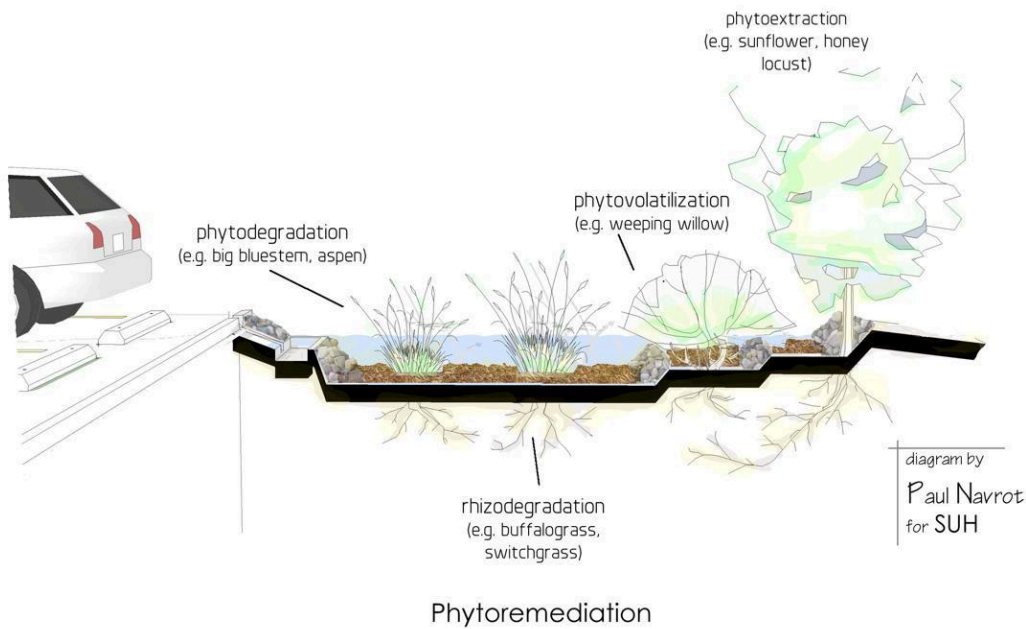
💣 **FACT:**

👉 **99% atmospheric sulfur = human activities**

⚠️ **MCQ:**

- **Cause of acid rain → SO_2**

PHYTOREMEDIATION (IMPORTANT)



✓ Definition:

👉 Use of plants to remove pollutants

⚙️ Processes involved:

- Sedimentation (S)
- Precipitation (P)
- Adsorption (Ad)

🔥 Application:

- Wastewater treatment using constructed wetlands

 MCQ:

- Phytoremediation uses → plants (not bacteria alone)

 FINAL ULTRA-REVISION (COMPLETE SYLLABUS)

 MOST IMPORTANT POINTS (LAST LOOK)

- BOD_5 → 5 days, 20°C
- High BOD = polluted water
- $COD > BOD$
- DO must not fall below 2 mg/L
- Pathogens:
 - Virus → Hepatitis, Rotavirus
 - Bacteria → Cholera
 - Protozoa → Giardia
 - Helminths → worms
- Cycles:
 - Carbon → lithosphere largest
 - Nitrogen → fixation required
 - Phosphorus → NO gas phase
 - Sulfur → acid rain
- Nitrogen:
 - Fixation → NH_3
 - Nitrification → NO_3^-
 - Denitrification → N_2
 - N_2O → 265–298× CO_2
- Treatment:
 - Composting → aerobic
 - Anaerobic → biogas
 - MFC → electricity
- Eutrophication:
 - Cause → N, P
 - Effect → algal bloom + fish death
- Limits:
 - BOD → 30 mg/L
 - COD → 250 mg/L
 - Water stress → 40%